GAS GUARD-INTELLIGENT LPG LEAK DETECTION AND ALERT SYSTEM

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Abstract— LPG leakage can occur suddenly during the normal use of LPG cylinders in homes, hotels, restaurants and other places which lead to serious accidents if it is not detected and controlled at the right time.In this reserch work, GasGuard has been designed for real-time monitoring and automatic safety response. The proposed model uses MQ-2 gas sensor to detect the LPG level and checks the condition of the sensor before checking the detected gas concentration. This model provides a safe or warning status according to the severity of the leakage. When LPG level reaches the dangerous condition, the regulator automatically turns OFF to stop the gas supply, while an exhaust fan turns ON to remove the accumulated gas. GasGuard is a practical LPG safety system that combines detection, severity analysis, automatic control and alert mechanisms.

Keywords—— LPG Leakage Detection, MQ-2 Gas Sensor, Sensor Health Monitoring, Gas Severity Analysis, Automatic Regulator Control, Exhaust Fan.

I. INTRODUCTION

LPG has emerged as a source of cooking fuel for various household uses and commercial purposes. However, there is the possibility of fire, losses of property, and even human casualties due to LPG leakage. All the current LPG leakage detection systems use gas sensors with manual safety intervention after gas leakages have been detected using gas sensors. GasGuard is the intelligent Internet of Things based LPG leakage detection and safety system that uses gas sensor and ESP32 development board to detect the status of the gas sensor and classify it as safe, warning, dangerous, and critical. The features of GasGuard include LPG regulator valve control, exhaust fan control, and LED. Therefore, GasGuard is the intelligent LPG leakage detection system that ensures LPG safety, detects and classifies the gas leakage, and controls the LPG valves and exhaust fans with safety alarms.

II. LITERATURE SURVEY

Gupta et al. [1] proposed a system for leakage detection and automatic shut-off of LPG using Arduino Uno and MQ-2 gas sensor . The system detects the LPG leakage, gives an initial alarm, and turns off the gas pipeline using a solenoid valve in the case of a hazardous leakage of LPG. The system also has an LCD screen to display the status of the gas.

Gavaskar et al. [2] proposed an IoT based LPG leakage detection, alarm and auto-exhaust system. This system is mainly designed for the detection of LPG leakage and activating an exhaust system for eliminating the leaked gas.

Altamash et al. [3] proposed a smart IoT-based fire and gas leakage detection system using ESP32, sensor fusion, Edge AI and ESP32-CAM.This system uses gas, flame and temperature sensors for improved detection and also makes use of visual confirmation to avoid false alerts. The proposed system is capable of making distinction between actual fire or gas leakage and other factors like smoke.

Proposed by Selvi and Ab Ghafar et al. [4], there is an IoT-based LPG leakage detection and monitoring system that uses MQ-2, NodeMCU ESP8266, and Blynk, which can detect the gas leakage and monitor the weight of the cylinder.

Another proposal for LPG leakage detection was made by Susilo and Rakhmawati et al. [5], in which ESP32 based fire and LPG leakage detection system using MQ-2 sensors and Telegram was used.

Anjum et al. [6] have suggested another LPG leakage detection system using an MQ-2 sensor and microcontroller. This system offers alerts using buzzer and LED and also shows the status of the gas.

Pamungkas et al. [7] suggested an LPG leakage detection system implemented via Arduino board and an MQ-2 sensor. The system utilizes a buzzer and blower fan to handle the leakage at first place, while the servo and spray foam mechanism ensure additional safety measures in case of high gas concentration. The system obtained the average reaction time of 9.56 seconds.

Sohal et al. [8] created an LPG leakage detection system based on Internet of Things technologies with the use of ESP32, MQ-2, DHT22, and LoRa sensors. The system provides local alerts via LEDs and a buzzer, while sending data on the leakage to a mobile application for remote monitoring purposes. Testing revealed reliable detection and notifications within the 10-30 m radius.

Asha et al. [9] suggested an IoT-based system to monitor the LPG/CNG gas leakage and the weight of the cylinders. The system implements Arduino Uno, gas sensor, load cell, LCD, buzzer, and GSM technologies and detects gas leakage, monitors weight of the cylinder, shows the real-time values and sends SMS notifications to users.

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Novitaningrum et al. [10] created an IoT-based system to mitigate LPG leakage using an MQ-2 sensor, ESP8266, LEDs, a buzzer and Blynk application. The system is able to detect the leakage of the LPG gas depending on certain threshold values and send alerts both locally and remotely via Blynk application. The system detected LPG within the 1-16 cmrange.

Table 1 Study on Gas leakage for detecting safety

Existing research work	Detection Method	Auto Safety Action	Sensor Health Check
Tukkoji &kumar[1]	MQ-2 (analog)	Not found	Not found
Altamesh et al.[3]	MQ-2 + Sensor Fusion(ML)	Not found	Not found
Tamil selvi& Ab Ghafar[5]	MQ-2(analog)	Not found	Not found
Pamungkas et.al.[7]	MQ-2 (analog)	Found	Not found
Duggirala et.al.[9]	Gas sensor(analog)	Not found	No found

III. RESEARCH GAP

In light of the review of the above-discussed systems, it is evident that there have been different types of gas sensors, microcontrollers, communication systems, alarm devices, exhaust fans and automatic cutoff mechanisms applied in detection of LPG leakage. Nonetheless, almost all of the above works have noted several challenges. First, a lot of the systems reviewed above employing gas sensor-based threshold detection systems experience challenges related to their accuracy in terms of variations in the sensors and the environment. It is not emphasized the need for health check of the sensors prior to the detection of the gases. Automatic regulation of the cutoff and exhaust fan together with the necessity of carrying out severity analysis is not emphasized either. Internet of Things based systems employ internet connectivity for remote control and monitoring purposes. This might be limited in case of failure in network. The proposed GasGuard overcomes most of these problems through emphasizing on the need for health check of the sensors, real-time detection of LPG leakage, severity classification, and automatic cutoff of the regulators and exhaust fan activation through an economical system with ESP32 as the microcontroller. The system is aimed at delivering a faster and more automated solution to the problem of LPG gas leakage than the previous works.

IV. PROPOSED METHODOLOGY

A.OVERVIEW OF GASGUARD

GasGuard is a very intelligent LPG Leakage Monitoring and Safety System which will help to monitor LPG Leakage and react accordingly. GasGuard is equipped with an MQ-2 Gas Sensor and ESP32 to monitor the environment for leakage. The GasSensor is highly reliable and its function is verified before taking any readings. The ESP32 receives the input from the Gas Sensor and analyzes the level of the gas and the severity of the problem. After analysis of the situation, the status will be displayed by the LED and displayed on OLED/LCD screen. If there is any danger because of the leakage, then LPG regulator will be switched off and Exhaust Fan will be activated.

B. SYSTEM ARCHITECTURE

The system architecture for the proposed model is shown in Figure 1.

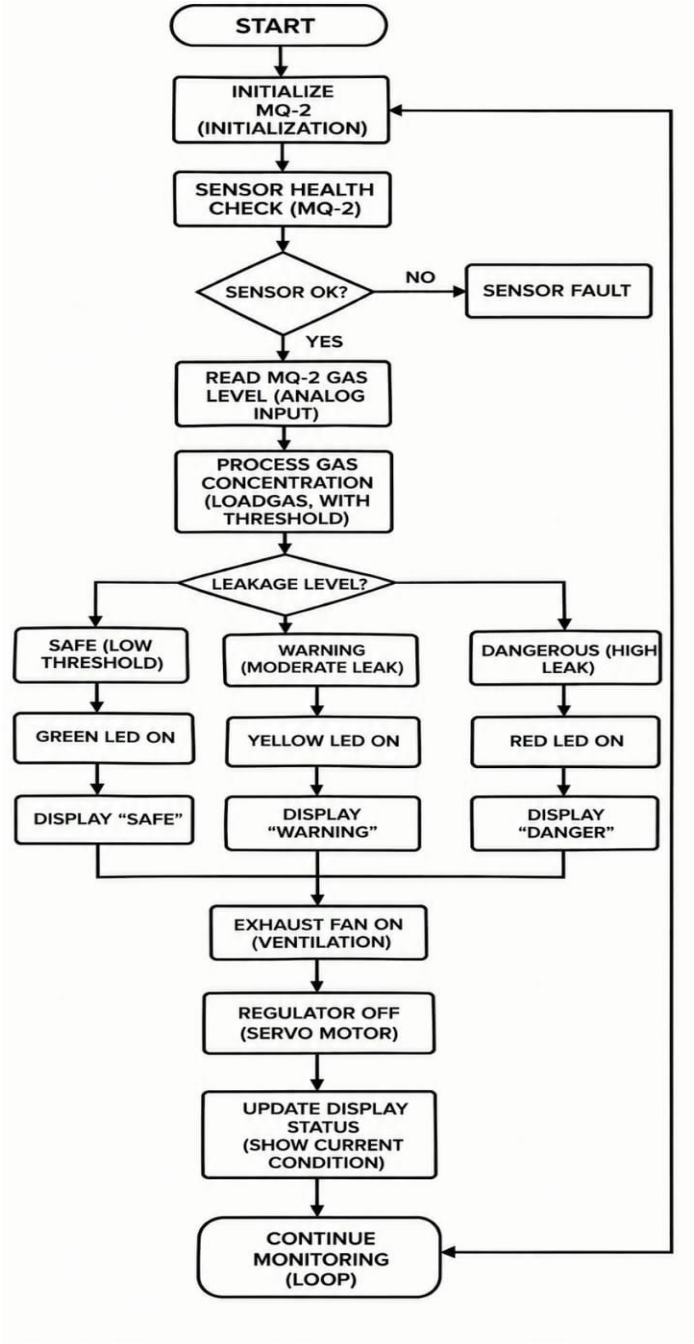


Figure 1. System Architecture

The suggested system uses ESP32 to control the LPG leakage monitoring and safety functions. Initially, MQ-2 gas sensor

undergoes stability test before estimating the quantity of gas. The results are compared with threshold values and classified based on standard levels of safe, cautious and dangerous values of gas. The three classes are depicted through three different colored LEDs and this information is simultaneously displayed on the OLED display. In case higher levels of LPG leaks persist, exhaust fan is turned on using relay circuit for higher expulsion of the gas. The Servo motor closes the LPG valve automatically for the safety mechanism and the system continues to monitor and display the information on the OLED screen. The entire experimental setup of this research work is depicted in Figure 2.

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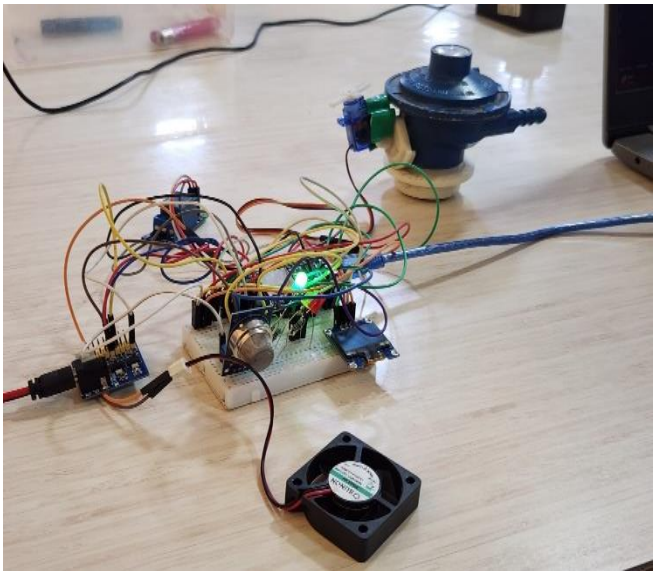


Figure 2. Experimental setup (Connecting IoT devices)

C. SYSTEM WORKFLOW

In addition to that, the test was also carried out on the GasGuard where the source of gas production used is the gas lighter which produces consistent gas that helps to test whether the MQ-2 sensor is working properly. The thresholds for the output of the sensors have been set to define the four possible states which are: Safe, Warning, Danger and Critical. If the state detected is in the safe state, then the green LED lights up. On the other hand, if the state is in the warning state, then the yellow LED lights up whereas in Danger and Critical states, the red LED lights up. The information will be presented using the OLED display device. In a case where the state of the gas is dangerous and critical, the exhaust fan will operate creating proper ventilation. However, in case of automatic shut off of the state because of the leaking gas, then the regulator shut off will work.

V.SYSTEM IMPLEMENTATION

A. USED IoT DEVICES

B. ESP-32 DEVELOPMENT BOARD

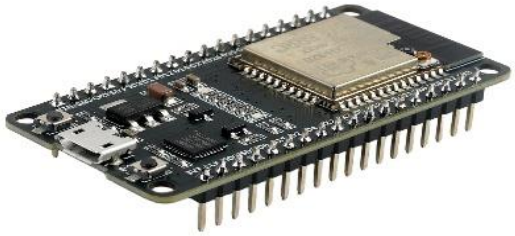


Figure 3.ESP32 DEVELOPMENT BOARD

The ESP32 development board which is shown in Figure 2 represents the central processing unit of the GasGuard and is responsible for receiving the information from the MQ-2 sensor about the LPG concentration level. It interprets the information according to established threshold values and determines the system state of whether Safe, Warning, Danger, or Critical. Moreover, it is the ESP32 development board that is responsible for controlling the LED diodes and OLED display, switching off the regulator, and turning on the exhaust fan to ensure the safe operation of the system under dangerous leakage conditions.

C. MQ-2 GAS SENSOR



Figure 4.MQ-2 GAS SENSOR

The MQ-2 gas sensor is the main sensor part of the GasGuard shown in Figure 4. system that detects the LPG leakage. It detects the LPG gas presence and provides the output signal to the ESP32. The working principle of this sensor is based on threshold values that define sensor states determined by the processing unit to classify the situation as Safe, Warning, Danger, or Critical. Altogether, the MQ-2 gas sensor plays major role in the operation of the GasGuard system to ensure that LPG levels are monitored on a regular and efficient basis.

D. SERVO MOTOR



Figure 5 SERVO MOTOR

The Servo motor is shown in Figure 5. It is used to operate the automatic LPG regulator shutoff mechanism in the GasGuard system. When the ESP32 detects a dangerous or critical LPG leakage condition, it sends a control signal to the servo motor. The servo motor then rotates the regulator mechanism to turn OFF the LPG supply, thereby helping to prevent further leakage.

E. EXHAUST FAN



Figure 6 Exhaust Fan

The exhaust fan is shown in Figure 6.it shall be utilized to enhance the performance of the ventilator in cases when the system senses a high LPG leakage. A relay module will be used to interface the ESP32 and the exhaust fan. Whenever the gas level that is detected by the system reaches any of the danger or critical gas level, ESP32 emits a control signal to the relay. This will energize the latter and consequently, turns the exhaust fan ON to expel the undesired gas in the targeted location offering automated safety mechanism.

F. LEDs

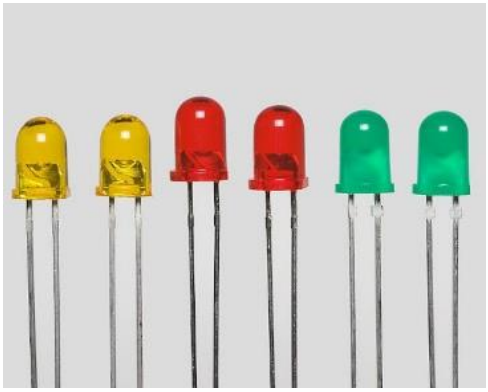


Figure 7 LEDs

The use of Light Emitting Diodes (LEDs) is shown in Figure 7. It is implemented in the GasGuard system to indicate the LPG leakage condition detected by the system. There are 3 colors that are used to show the indication which are the Green, Yellow, and Red colors. All these colors were used to indicate the level of safety condition of the LPG. For instance, a green color is used to indicate that the status of LPG is Safe while a yellow color is implemented to indicate that the status of LPG is Warning. On the contrary, a red color is used to indicate that either the Danger or Critical condition of the LPG is happening. The ESP32 is implemented in the system to control the LED depending on the level of gas detected by the MQ-2 sensor. It will help the user to be aware of the state of the LPG condition detected based on the color that is indicated during the operation.

F.OLED



Figure 8 OLED Display

The OLED display picture is shown in Figure 8. It is used in the GasGuard system to provide the information of the detected LPG condition. The OLED display is connected to the ESP32 and is used to display the current status according to the MQ-2 sensor readings. The OLED display displays the status of Safe mode, Warning mode, Danger mode, and Critical mode to provide the indication to the user. It provides the detected condition of the LPG leakage to the user that increases the user interface of the system.

G. LPG REGULATOR



Figure 9 LPG REGULATOR

The LPG regulator is shown in Figure 9. It is used to control the flow of LPG from the cylinder to the connected appliance. In the GasGuard system the regulator is in conjunction with the automatic shutoff mechanism operated by the Servo motor . Upon the detection of dangerous or critical leakage condition by the MQ-2 sensor, the ESP32 sends a control signal to operate the motor through the motor driver. The regulator is turned OFF to cut off the LPG supply which helps in preventing further leakage during hazardous conditions.

G. 1-CHANNEL RELAY MODULE

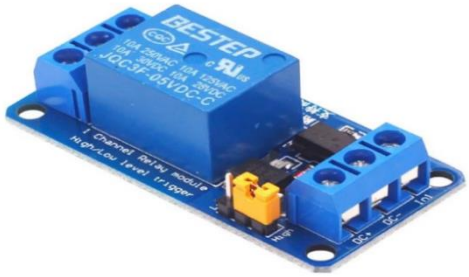


Figure 10 1-CHANNEL RELAY MODULE

The 1-channel relay module is shown in Figure 10. It is used as a switching interface between the ESP32 and the exhaust fan. Thus , when the MQ-2 sesor detects a danger or critical LPG leakage, the microcontroller sends a control signal to the relay module. Consequently, the latter switches the exhaust fan ON, providing automatic ventilation.

B.SOFTWARE USED

A.Arduino IDE

Arduino IDE is used to write, compile and upload the control program on ESP32 development board.

B. Embedded C/C++

Embedded C/C++ is used for programming ESP32 board and implementing the LPG detection, threshold analysis and safety control algorithm.

C.ESP32 Board Package

ESP32 board package is included in the Arduino IDE, for compiling and uploading the program specific for the ESP32 controller.

D. Required Libraries

Libraries are used to interface the MQ-2 sensor, OLED display, LEDs, relay and motor-control devices with ESP32.

E.Serial Monitor

Arduino serial monitoring tool is used while testing for observing the sensor values and verifying the defined threshold values and reactions of the system.

C.RESULTS AND DISCUSSION

A comparison of the existing LPG leakage detection systems with the proposed GasGuard system is shown in Table 2. It reveals that most of the systems focus on only detection and monitoring with limited response mechanism such as cloud backup or GSM alert. The systems also do not involve sensor health indicator.

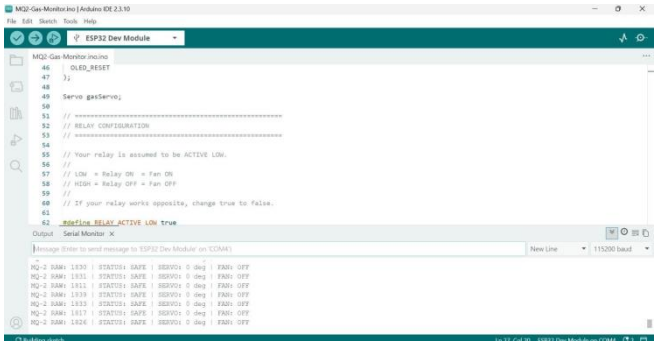


Figure 11 INDICATING STATUS SAFE

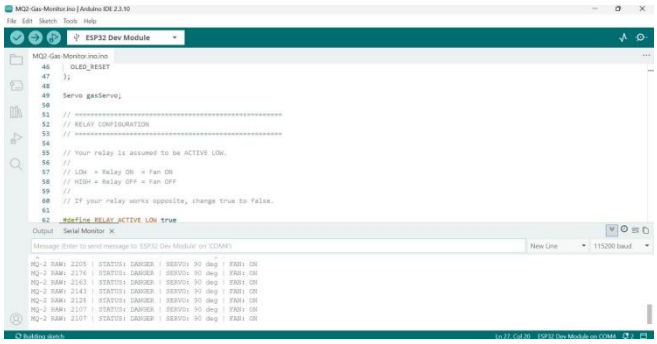


Figure 12 INDICATING STATUS DANGER

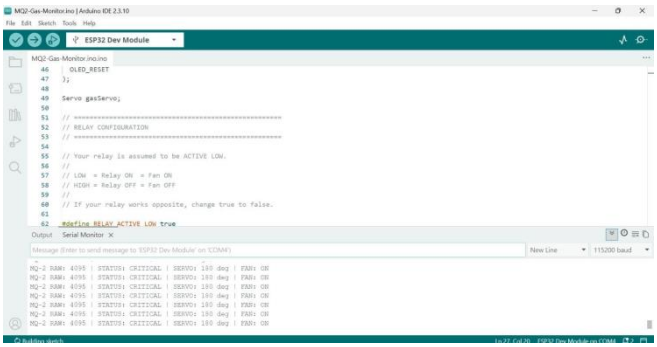


Figure 13 INDICATING STATUS CRITICAL

Table 2 Feature comparison table on both existing and proposed system

S.NO	Features Comparison		
	AUTHOR	EXISTING	PROPOSED
1	[1]	It detects and displays only gas level , no auto action and uses Arduino IDE	It has auto regulator cutoff and exhaust fan to remove accumulated gas
2	[3]	It uses ML+ camera and no cutoff, cloud dependent	Auto cutoff ,works without cloud
3	[5]	Weight monitoring, no auto action	OLED display,auto action
4	[7]	Auto action and no sensor health check	Sensor health check and auto action
5	[9]	GSM alerts only and no auto action	Local auto action and no GSM module

The GasGuard system on the other hand is a combined system of sensor health indicator, LPG detection, OLED display, exhaust-fan activation and regulator shutoff which is able to perform local and automatic response to gas leakage without the need for cloud backup or GSM connection.

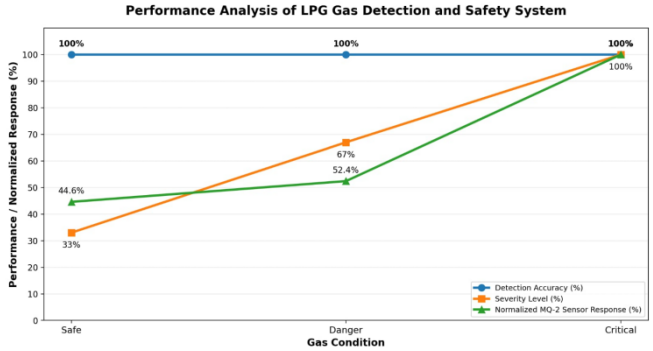


Figure 14 PERFORMANCE ANALYSIS

The Graph shows the system performance under safe, danger and critical conditions. Detection accuracy remains 100% for all conditions. Severity increases from 33% to 67% and 100%, while the MQ-2 sensor response rises from 44.6% to 52.4% and 100%. These results show that the system can effectively detect and classify different LPG leakage levels.

VI.CONCLUSION AND FUTURE WORK

The GasGuard concept is an excellent one, for it features both the ability to detect the occurrence of a leak and to respond to it appropriately. The prototype of the gas control system proves that it has gas sensing and status indicator, ventilation and regulator control, as it is responsive and efficient. Therefore, it is possible to suggest that the system would be further developed and refined in the future to account for greater efficiency. Firstly, the system could be improved with the use of a load cell to weigh the gas cylinder and notify the user that it needs to be refilled. Secondly, the addition of a buzzer or a siren to the system would benefit the system as it would allow it to raise an alarm over a greater distance. Lastly, it would be beneficial to test the system under a variety of conditions to ensure its reliability and practicality.

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